

hw5-cardinality-functions-relations: Sample Solutions

CSE20W26

In this assignment, you will work with cardinality, functions, and binary relations and prove properties about them, practicing proof strategies and applications.

Relevant class material: Weeks 8,9,10.

You will submit this assignment via Gradescope (<https://www.gradescope.com>) in the assignment called “hw5-cardinality-functions-relations”.

In your proofs and disproofs of statements below, justify each step by reference to a component of the following proof strategies we have discussed so far, and/or to relevant definitions and calculations.

Assigned questions

1. Binary relations. In the review quiz, we considered the binary relation on $\mathbb{Z}^+$ defined by

$$\{(a, b) \mid \exists c \in \mathbb{Z}(b = ac)\}$$

Let's call that relation R_1 . Consider the following other binary relations on $\mathbb{Z}^+$:

$$R_2 = \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid \gcd(a, b) = 1\}$$

$$R_3 = \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid a + b \leq 2024\}$$

Sample response that can be used as reference for the detail expected in your answer: To prove that R_1 is not symmetric we need to find a counterexample to

$$\forall x \in \mathbb{Z}^+ \forall y \in \mathbb{Z}^+ ((x, y) \in R_1 \rightarrow (y, x) \in R_1)$$

Consider $x = 2$ and $y = 4$. Then $(x, y) \in R_1$ because $\exists c \in \mathbb{Z}(y = xc)$ is witnessed by the integer $c = 2$, since $4 = 2 \cdot 2$. However, $\exists c \in \mathbb{Z}(x = yc)$ is not true: to prove this, we consider an arbitrary integer c and consider the two (exhaustive) cases $c \geq 1$ and $c < 1$. In case 1, assume $c \geq 1$, and then $yc = 4c \geq 4 > 3$ so $yc \neq 2 = x$. In case 2, assume $c < 1$ so $c \leq 0$, and then $yc = 4c \leq 0$ so is not equal to 2, which is x . Thus, we've found positive integers x and y where $(x, y) \in R_1$ and $(y, x) \notin R_1$ so R_1 is not symmetric.

- (a) Give example positive integers that are related according to each of these three relations. In other words, give example values (a_1, b_1) , (a_2, b_2) , and (a_3, b_3) such that $(a_i, b_i) \in R_i$ for each $i = 1, 2, 3$.

Sample Solution: Let $a_1 = 1, b_1 = 2$ (both positive integers). $(a_1, b_1) \in R_1$ as $c = 2$ witnesses $\exists c \in \mathbb{Z} b_1 = a_1 \cdot c$.

Let $a_2 = 3, b_2 = 5$ (both positive integers). The positive factors of a_2 form the set $\{1, 3\}$, and the positive factors of b_2 form the set $\{1, 5\}$, so $\gcd((a_2, b_2)) = 1$ and thus $(a_2, b_2) \in R_2$.

Let $a_3 = 1, b_3 = 1$ (both positive integers). $a_3 + b_3 = 2 \leq 2024$, so $(a_3, b_3) \in R_3$.

- (b) Prove that R_2 is not transitive.

Solution: To prove that R_2 is not transitive we need to find a counterexample to

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ \forall c \in \mathbb{Z}^+ \quad ((a, b) \in R_2 \wedge (b, c) \in R_2) \rightarrow (a, c) \in R_2$$

Consider $a = 2, b = 3, c = 4$ (all of which are positive integers). Then $(a, b) \in R_2$ as $\gcd((2, 3)) = 1$, and $(b, c) \in R_2$ as $\gcd((3, 4)) = 1$. However, $(a, c) \notin R_2$ as $\gcd((2, 4)) = 2 \neq 1$. Hence, we have found positive integers a, b, c where $(a, b) \in R_2$ and $(b, c) \in R_2$ but $(a, c) \notin R_2$, so R_2 is not transitive.

- (c) Prove that R_3 is not reflexive.

Solution: To prove that R_3 is not reflexive we need to find a counterexample to

$$\forall a \in \mathbb{Z}^+ \quad (a, a) \in R_3$$

Consider $a = 2000$, $a + a = 4000 > 2024$, so $(a, a) \notin R_3$. Hence, we have found positive integer a where $(a, a) \notin R_3$, so R_3 is not reflexive.

- (d) Prove that R_2 is symmetric.

Solution: To prove that R_2 is symmetric we need to prove

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ \quad (a, b) \in R_2 \rightarrow (b, a) \in R_2$$

Let a, b be arbitrary positive integers. Towards a direct proof, assume $(a, b) \in R_2$, which implies $\gcd((a, b)) = 1$. By the definition of $\gcd$ (because the set of common divisors of two integers can be thought of as the intersection of the separate sets of divisors, and set intersection is commutative), $\gcd((a, b)) = \gcd((b, a))$. Hence, $\gcd((b, a)) = 1$ and $(b, a) \in R_2$.

- (e) Prove that R_3 is not anti-symmetric.

Solution: To prove that R_3 is not anti-symmetric we need to find a counterexample to

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ \quad ((a, b) \in R_3 \wedge (b, a) \in R_3) \rightarrow a = b$$

Consider $a = 1, b = 2$. Then $a + b = b + a = 3 \leq 2024$, which implies $(a, b) \in R_3 \wedge (b, a) \in R_3$. However, $a \neq b$. Hence, we have found positive integers a, b, c where $((a, b) \in R_3 \wedge (b, a) \in R_3) \rightarrow a = b$ is false, so R_3 is not anti-symmetric.

- (f) Give an example relation on $\mathbb{Z}^+$ that is both symmetric and anti-symmetric. Briefly justify your example.

Sample solution 1: The identity relation $R = \{(a, b) \in \mathbb{Z}^+ \times \mathbb{Z}^+ \mid a = b\}$ is both symmetric and anti-symmetric. It is symmetric because equals is symmetric, so $\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ (a = b \rightarrow b = a)$. It is anti-symmetric because, if we consider arbitrary positive integers a and b and assume that $(a, b) \in R$ and $(b, a) \in R$, the definition of R guarantees that $a = b$ as required.

Sample solution 2: The empty relation $R = \emptyset$ is both symmetric and anti-symmetric. The universal claim $\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ ((a, b) \in R \rightarrow (b, a) \in R)$ evaluates to true because the hypothesis is always false. Similarly $\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+ ((a, b) \in R \wedge (b, a) \in R \rightarrow a = b)$ also evaluates to true because the hypothesis is always false. Hence, R is both symmetric and anti-symmetric.

2. Equivalence relations. Recall that we say a is **congruent to b mod n** means $(a, b) \in R_{(\text{mod } n)}$, that is $(a \text{ mod } n = b \text{ mod } n)$. A common notation is to write this as $a \equiv b(\text{mod } n)$.

A **modular inverse** of an integer x relative to modulus n (where n is a positive number and x is an integer between 0 and $n - 1$, inclusive) is defined to be an integer y between 0 and $n - 1$ (inclusive) such that $xy \equiv 1(\text{mod } n)$.

- (a) Fill in the two multiplication tables below. In the left table, each cell in row labelled a and column labelled b is an integer p in the set $\{0, 1, 2, 3, 4\}$ satisfying $ab \equiv p(\text{mod } 5)$. In the right table, each cell in row labelled a and column labelled b is an integer p in the set $\{0, 1, 2, 3, 4, 5\}$ satisfying $ab \equiv p(\text{mod } 6)$. **Solution:**

	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

- (b) Does each integer x between 1 and 4 (inclusive) have a modular inverse relative to modulus 5? Does each integer y between 1 and 5 (inclusive) have a modular inverse relative to modulus 6? Justify your answers using appropriate proof strategies and definitions.

Sample solution: The universal claim is true about modular inverses relative to modulus 5, but not about modular inverses relative to modulus 6. Reading across the rows in the multiplication table for modulus 5, we see that the modular inverse of 1 is 1, the modular inverse of 2 is 3, the modular inverse of 3 is 2, and the modular inverse of 4 is 4. A counterexample to the existence of modular inverses relative to modulus 6 is 2, an integer between 1 and 5 and every entry in the row of the multiplication table for it isn't 1.

- (c) Is there an example of integers x_1, x_2 between 1 and 4 (inclusive) so that $x_1x_2 \equiv 0(\text{mod } 5)$?
 Is there an example of integers y_1, y_2 between 1 and 5 (inclusive) so that $y_1y_2 \equiv 0(\text{mod } 6)$?
 Justify your answers using appropriate proof strategies and definitions.

Sample solution: There are no integers x_1, x_2 between 1 and 4 (inclusive) so that $x_1x_2 \equiv 0(\text{mod } 5)$ because all the entries in the left multiplication table (aside from the top row and left column) are nonzero. Example y_1, y_2 between 1 and 5 (inclusive) so that $y_1y_2 \equiv 0(\text{mod } 6)$ are $y_1 = 2$ and $y_2 = 3$. To justify that these are witnesses using the definition of remainders, we calculate: $y_1y_2 = 2 \cdot 3 = 6 = 1 \cdot 6 + 0$ so $y_1y_2 \equiv 0(\text{mod } 6)$.

3. Partial orders. Recall the recursive definition of the set of linked lists of natural numbers (from class)

Basis Step: $[] \in L$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$, then $(n, l) \in L$

and the definition of the function which gives the length of a linked list of natural numbers
 $length : L \rightarrow \mathbb{N}$

Basis Step: $length([]) = 0$

Recursive Step: If $l \in L$ and $n \in \mathbb{N}$, then $length((n, l)) = 1 + length(l)$

For this question, we'll restrict our attention to just linked lists with data in the set $\{0, 1\}$ which have length 0, 1, or 2. Let's call this restricted set of lists L' .

Note: L' has 7 distinct elements.

Define two different partial orders with domain L' . For each of the partial orders, you can specify its definition by listing the set of ordered pairs with roster method, giving a set builder definition, giving a recursive definition, or using a clear and precise English description of which lists are related to one another. Then, draw the Hasse diagram for each of the two partial orders you defined.

Sample solution 1: The simplest partial order is $R = \{(a, a) \mid a \in L'\}$. It is reflexive as $(a, a) \in R$ for each element a in L' (we can test this by exhaustion). It is anti-symmetric because, for arbitrary a and b in L' , if we assume that $(a, b) \in R \wedge (b, a) \in R$, that guarantees (in particular) that $(a, b) \in R$, which by definition means $a = b$, as desired. It is transitive because for arbitrary a, b, c in L' , if $((a, b) \in R \wedge (b, c) \in R)$ the definition of R gives $a = b \wedge b = c$ and since $=$ is transitive, we get $a = c$, or in other words, $(a, c) \in R$ (as required). Hence, R is a partial order.

The Hasse diagram would have no edges (as we omit self-loops):

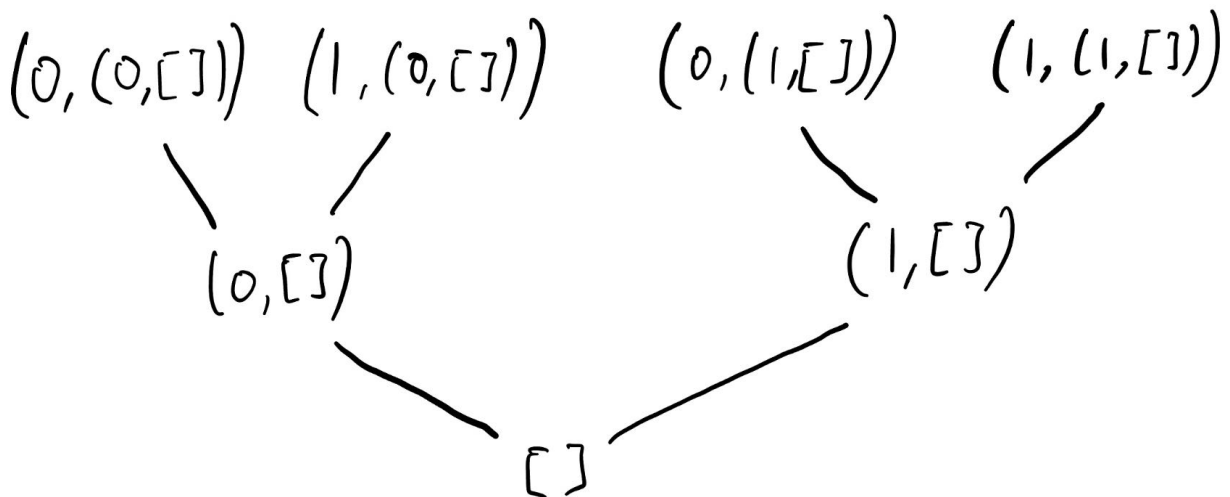
$[]$ $(0, [])$ $(1, [])$ $(0, (1, []))$ $(1, (1, []))$ $(0, [])$ $(1, [])$ $[]$

Sample Solution 2: Another partial order can be defined by

$R_2 = \{(a, b) \in L' \times L' \mid \text{we can change } a \text{ to } b \text{ by prepending data to } a \text{ non-negative many times}\}$

It is reflexive as, for arbitrary a in L' , we can change a to a by prepending data zero many times, which means $(a, a) \in R_2$. To formally prove that R_2 is anti-symmetric, we could prove a lemma that, for all $(a, b) \in R_2$, $\text{length}(a) \leq \text{length}(b)$. Once we have that, we can show that, for arbitrary a and b in L' , if $(a, b) \in R_2 \wedge (b, a) \in R_2$, we $\text{length}(a) \leq \text{length}(b)$ and $\text{length}(a) \geq \text{length}(b)$ so $\text{length}(a) = \text{length}(b)$ and the only way to have that while the relation holds is to have $a = b$. It is transitive because for arbitrary a, b, c in L' , if we assume that $((a, b) \in R_2 \wedge (b, c) \in R_2)$, that means we can change a to b and b to c by prepending, therefore we can also change a to c by combining those operations. So $(a, c) \in R_2$ as desired. Hence, R_2 is a partial order.

The Hasse diagram would look like:



4. Equivalence classes and partitions. Recall that in a movie recommendation system, each user's ratings of movies is represented as a n -tuple (with the positive integer n being the number of movies in the database), and each component of the n -tuple is an element of the collection $\{-1, 0, 1\}$. Assume there are five movies in the database, so that each user's ratings can be represented as a 5-tuple. We call Rt_5 the set of all ratings 5-tuples. Consider the binary relation on the set of all 5-tuples where each component of the 5-tuple is an element of the collection $\{-1, 0, 1\}$:

$$G = \{(u, v) \in Rt_5 \times Rt_5 \mid \text{the number of 1s in } u \text{ is the same as the number of 1s in } v\}$$

This is an equivalence relation (you do not need to prove this).

Recall that the **equivalence class** of an element $x \in X$ for an equivalence relation $\sim$ on the set X is the set $\{s \in X \mid (x, s) \in \sim\}$. We write this as $[x]_{\sim}$.

- (a) Find a ratings 5-tuple v such that $[v]_G = \{v\}$. Justify your choice of v .

Solution: $[v]_G = \{v\}$ means that the only element in the equivalence relation is the element itself. If there are any elements other than 1 in the 5-tuple v , another element in that equivalence relation could have the same 5-tuple but swap out one non-1 number element with the other number not equal to 1. For example, if we have $v = (1, 1, 1, 1, 0)$, the tuple $v' = (1, 1, 1, 1, -1)$ would also be in the equivalence relation set and v cannot be the only element in $[v]_G$. Hence, the only possible v is $(1, 1, 1, 1, 1)$. For all $u \in Rt_5$, $(u, v) \in G$ means the number of 1s in u is also five, which guarantees that $u = (1, 1, 1, 1, 1)$. Hence, the only member in $[v]_G$ is v itself.

- (b) Find distinct ratings 5-tuples u_1, u_2 ($u_1 \neq u_2$) whose equivalence classes $[u_1]_G$ and $[u_2]_G$ have the same size.

Note: We know that elements of the domain of an equivalence relation that are related by the equivalence relation have the same equivalence class.

Sample solution: All we need is to find $u_1 \neq u_2$ such that $(u_1, u_2) \in G$. One example would be $u_1 = (1, 0, 0, 0, 0)$ and $u_2 = (0, 1, 0, 0, 0)$. They have the same number of 1s and thus $(u_1, u_2) \in G$. By the note above, $[u_1]_G = [u_2]_G$ and thus they have the same size.

- (c) Find distinct ratings 5-tuples w_1, w_2 ($w_1 \neq w_2$) whose equivalence classes $[w_1]_G$ and $[w_2]_G$ have different sizes.

Sample Solution: Let $w_1 = (1, 0, 0, 0, 0)$ and $w_2 = (1, 1, 1, 1, 1)$. From (a) we know $[w_2]_G = \{w_2\}$ has size 1, and from (b) we know that $(0, 1, 0, 0, 0)$ is another element in $[w_1]_G$ other than w_1 , which means $[w_1]_G$ has size at least 2. Hence, $[w_1]_G$ and $[w_2]_G$ have different sizes.

5. Modular exponentiation. Imagine you are playing the role of Alice in the Diffie Hellman key agreement (exchange) protocol. You and Bob have agreed to use the prime $p = 7$ and its primitive root $a = 3$. Your secret integer is $k_1 = 3$.

- (a) Calculate the number you send to Bob, $a^{k_1} \bmod p$. Use the modular exponentiation algorithm for the calculation. Include a trace of the algorithm in your solution.

Modular Exponentiation

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1  procedure modular_exponentiation( $b$ : integer;
2       $n = (a_{k-1}a_{k-2} \dots a_1a_0)_2$ ,  $m$ : positive integers)
3       $x := 1$ 
4       $power := b \bmod m$ 
5      for  $i := 0$  to  $k - 1$ 
6          if  $a_i = 1$  then  $x := (x \cdot power) \bmod m$ 
7           $power := (power \cdot power) \bmod m$ 
8      return  $x$  { $x$  equals  $b^n \bmod m$ }

```

Solution:

Trace for Modular Exponentiation(b=3, n=3 = (11) ₂ , m=7)				
	x	power	i	a _i
Initial Value	1	3	0	1
After 1 Iteration	3 (=1*3 mod 7)	2 (=3*3 mod 7)	1	1
After 2 Iteration	6 (=3*2 mod 7)	4 (=2*2 mod 7)		

Hence, $a^{k_1} \bmod p = 3^3 \bmod 7 = 6$.

- (b) Bob sends you the number 5. Compute your shared key, $(a^{k_2})^{k_1} \bmod p$. Hint: $a^{k_2} \bmod p$ is what Bob sent you. Include all relevant calculations, annotated with explanations, for full credit.

Solution: Since exponentiation is just a sequence of multiplication and by the multiplication arithmetic in modular, we have $a^b \bmod p = (a \bmod p)^b \bmod p$. Hence,

$$\begin{aligned}
 & (a^{k_2})^{k_1} \bmod p \\
 &= (a^{k_2} \bmod p)^{k_1} \bmod p \\
 &= 5^{k_1} \bmod 7 \\
 &= 5^3 \bmod 7 \\
 &= 125 \bmod 7 \\
 &= 6
 \end{aligned}$$

- (c) What are some possible values for Bob's secret integer? What algorithm are you using to compute them?

Sample Solution: We want to find k_2 such that $3^{k_2} \bmod 7 = 5$. We can start off by trying few small values for k_2 and try to find a pattern.

$$\begin{aligned}
 3^1 \bmod 7 &= 3 \\
 3^2 \bmod 7 &= 3 \cdot 3 \bmod 7 = 2 \\
 3^3 \bmod 7 &= 2 \cdot 3 \bmod 7 = 6 \\
 3^4 \bmod 7 &= 6 \cdot 3 \bmod 7 = 18 \bmod 7 = 4 \\
 3^5 \bmod 7 &= 4 \cdot 3 \bmod 7 = 12 \bmod 7 = 5 \\
 3^6 \bmod 7 &= 5 \cdot 3 \bmod 7 = 15 \bmod 7 = 1 \\
 3^7 \bmod 7 &= 1 \cdot 3 \bmod 7 = 3
 \end{aligned}$$

Hence, one possible value for k_2 is 5, and the other values are elements of the set $\{5 + 6 * n \mid n \in \mathbb{N}\}$.

The proof is based on the observation that $3^6 \bmod 7 = 1$, which means $3^{5+6*n} \bmod 7 = (3^5) \cdot (3^6)^n \bmod 7 = (3^5 \bmod 7) \cdot ((3^6)^n \bmod 7) = 5 \cdot ((3^6 \bmod 7)^n \bmod 7) = 5 \cdot ((1^n) \bmod 7) = 5 \cdot 1 = 5$.